Cloud Delphi framework under asymmetric uncertainty — R scripts

Supplementary code for:

> Cruz-Ramírez, M. *Cloud Delphi framework under asymmetric uncertainty*. Croatian Operational Research Review (CRORR). DOI: 10.17535/crorr.2027.0013

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These scripts implement the **asymmetric Cloud Delphi** model. Each expert's judgment is a cloud with two entropies, a left one (`En_L`) and a right one (`En_R`). Hyper-entropy (`He`) combines the expert's self-reported confidence with inter-round volatility. The panel opinion is synthesized round by round, and formal stopping criteria decide when the Delphi process ends. You can use them with your own panel data; see *Using your own data* below.

Files

File	Purpose
`CloudDelphi.R`	**Main script.** Processes all Delphi rounds, applies the stopping criteria, prints the metrics of each round (Table 1 of the article) and draws the evolution of the clouds (Figure 2).
`Sensitivity.R`	One-at-a-time sensitivity analysis of the five design constants (ε, ξ, 18, 2, 3), each perturbed by ±20% and ±50%.
`sensitivity_results.csv`	Output of `Sensitivity.R` for the panel in the article.
`Comparison.R`	Compares the asymmetric model with the classic symmetric Cloud Delphi baseline.
`AnisotropyAnalysis.R`	Reanalysis of the published data of Cruz & Cables (2022) (N = 56) on individual cognitive anisotropy.
`Pseudocode.pdf`	Pseudocode of the two core algorithms (one round, and the multi-round loop).

Requirements

- R version 4.0 or later (tested with R 4.5.2).
- No additional packages: everything uses base R (`stats`, `graphics`, `grDevices`).

Quick start

```
source("CloudDelphi.R")
```

This reproduces the case study of the article (N = 7 experts, 3 rounds). It does three things:

1. It prints the metrics of each round to the console, in the same layout as Table 1 of the article.
2. It reports the round in which the process stops, for example `[SYSTEM] Disjunction reached in Round 3. Evolution halted.`
3. It saves `CloudDelphi_Evolution_Analysis.eps` and `.pdf` to your `Documents` folder (Figure 2 of the article).

A fixed seed (`set.seed(2026)`) makes the random cloud drops reproducible.

Input data format

Each round is one numeric vector with **4 values per expert**, in this order:

`x_min, x_max, Ex, C`

Value	Meaning	Constraint
`x_min`	minimum forecast	`x_min <= Ex`
`x_max`	maximum forecast	`Ex <= x_max`
`Ex`	most probable forecast (cloud crest)	inside `[x_min, x_max]`
`C`	self-reported confidence	in `[0, 1]`

About the confidence scale. In the article's experiment, confidence was rated from 0 to 100 and divided by 100. The article recommends the simpler eleven-point scale {0.0, 0.1, ..., 1.0}, because coarsening to it did not change the results in any relevant way. Either scale works, as long as `C` ends up in [0, 1]. Note that `C = 0` gives that expert a weight of exactly zero in that round.

Example with 3 experts in one round:

```
Round1_Data <- c(15, 75, 20, 0.80,    # expert 1
                 10, 60, 52, 0.10,    # expert 2
                 30, 90, 35, 0.70)    # expert 3
```

The experts must appear **in the same order in every round**, because hyper-entropy compares each expert's answers with that same expert's answers in the previous round.

Using your own data

1. Open `CloudDelphi.R` and go to the section **EXECUTION PIPELINE** at the end.
2. Replace `Round1_Data`, `Round2_Data`, ... with your own vectors, using the format above.
3. List the rounds in `DelphiStudyRounds`:

   ```
   DelphiStudyRounds <- list(Round_1 = Round1_Data, Round_2 = Round2_Data)
   ```

4. If necessary, adjust the parameters in the **GLOBAL CONFIGURATION BLOCK** at the top:

Parameter	Default	Meaning
`epsilon`	0.01	numerical stabilizer (anisotropy index, weights, ΔEn)
`xi`	0.001	minimum split-entropy
`tau`	0.50	stopping threshold for the relative variation of the synthetic entropy
`theta`	0.20	stopping threshold for the variation of the uncertainty index `I_s`
`R_max`	4	maximum number of rounds
`Discourse_Min`, `Discourse_Max`	0, 100	universe of discourse (x-axis of the plots)
`Drops_Expert`, `Drops_Synthetic`	1200, 2500	number of drops drawn per cloud

The values of `tau`, `theta` and `R_max` depend on each study and should be set by the coordinating group.

To change where the figures are saved, pass `OutputDir`:

```
CloudDelphi_Evolution(RoundsList = DelphiStudyRounds, OutputDir = "C:/my/folder")
```

Output

For each round, the console shows one matrix with a column per expert (`Exp_1`, …) and a column `Exp_s` for the panel's synthetic cloud:

Row	Meaning
`Ex_Prob`	expectation (crest)
`En_Left`, `En_Right`	left and right split-entropies
`He_Hyper`	hyper-entropy
`Weight`	normalized expert weight (`Exp_s` = 1)
`Delta_En`	relative change of the mean entropy with respect to the previous round (each expert relative to their own previous value)
`Unc_Index`	uncertainty index `He / mean entropy`

The function returns the same matrices as a list (`Delphi_Outputs`).

Model summary

- **Split-entropies:** `En_L = (Ex − x_min)(2 − C)/3`, `En_R = (x_max − Ex)(2 − C)/3`, each bounded below by `xi`.
- **Initial hyper-entropy:** `He = (En_L + En_R) / (2·K)`, with `K = max((3 + 3C)(1 − A/2), 2)` and anisotropy index `A = |En_L − En_R| / (En_L + En_R + ε)`.
- **Later rounds:** an inter-round volatility term `Ω = (|Δx_min| + |Δx_max| + |ΔEx|)/18` is added to `He`.
- **Weights:** proportional to `C`, and inversely proportional to the expert's relative distance from the panel mean plus their entropy and hyper-entropy.
- **Cloud drops:** split double randomization. The side is chosen with probability `En_L/(En_L + En_R)` or `En_R/(En_L + En_R)`, the entropy of that side is perturbed with `He`, and the drop is placed on that side of `Ex`.
- **Stopping:** the process stops when the synthetic entropy stabilizes (`ΔEn < tau`), the uncertainty index stabilizes (`|ΔI_s| < theta`), or `R_max` is reached.

See `Pseudocode.pdf` and the article for the full formulation.

Other scripts

- **`Sensitivity.R`.** Edit section *1. RAW PANEL DATA* to use other data (the format is a list with `xmin`, `xmax`, `Ex`, `C` per round). It writes `sensitivity_results.csv` to the working directory (`getwd()`). In the CSV, `stopping_round = NA` means that the process did not stop within the available rounds.
- **`Comparison.R`.** Prints the percentage differences in entropy and hyper-entropy between the symmetric baseline and the asymmetric model for Rounds 1 and 2. In Round 3 every expert reported a symmetric range, so both models coincide.
- **`AnisotropyAnalysis.R`.** Uses the data of Cruz & Cables (2022), *Investigación Operacional*, 43(2), 102–119. Three experts per round reported a most-likely value outside their own range; they are retained as published.

Version notes

Compared with the first version of this repository, this version:

- computes `Delta_En` for each expert relative to that expert's own previous entropy (it previously used the maximum over the panel);
- implements the split drop generator described in the article (Eq. 8) and fixes the random seed;
- runs `Sensitivity.R` on the actual panel data and defines `I_s` as in the article (`He_s` divided by the mean synthetic entropy).

How to cite

Please cite the article above. If you use the code, please cite this Zenodo record as well.